Original Article

Synthesis of Thiazoles Using Amberlyst Resin as a Reusable Heterogeneous Catalyst

Popat M. Jadhav[1,2] | Anand V. Dhirbassi[1] | Sushama S. Kauthale[3] | Sunil U. Tekale[1,*]

[1]Department of Chemistry, Deogiri College, Chhatrapati Sambhajinagar, 431005, India
[2]Department of Chemistry, Shri Muktanand College, Gangapur, Dist: Chhatrapati Sambhajinagar, 431109, India
[3]Department of Chemistry, S.B.E.S. College of Science, Chhatrapati Sambhajinagar, 431001, India

Use your device to scan and read the article online

Citation P.M. Jadhav, A.V. Dhirbassi, S.S. Kauthale, S.U. Tekale, **Synthesis of Thiazoles Using Amberlyst Resin as a Reusable Heterogeneous Catalyst.** *Eurasian J. Sci. Technol.,* **2026**, 6(4), 294-298.

https://doi.org/10.48309/ejst.2026.581826.1318

CC BY

Article info:
Received: 2026-05-27
Accepted: 2026-08-20
Available Online: 2026-08-25
ID: EJST-2605-1318

Keywords:
Amberlyst resin; Aminothiazole; Phenacyl bromide; Thiourea

ABSTRACT

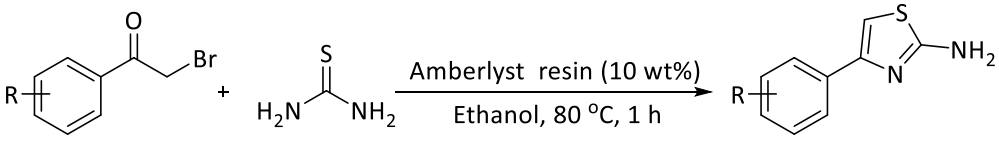
Thiazole is an important five-membered heterocyclic ring containing nitrogen and sulfur atoms, playing a crucial role in synthetic and medicinal chemistry. The ring has a unique structural ability for target binding and metabolic stability, which has resulted in its utility in several commercially marketed drugs. In the present work, a rapid and efficient method has been developed for the synthesis of substituted thiazole derivatives from various substituted phenacyl bromides and thiourea in refluxing ethanol. A series of 2-amino thiazoles was synthesized in good to excellent yield of the corresponding products using Amberlyst resin as a heterogeneous catalyst. The developed protocol offers excellent yield involving a reusable catalyst and an easy work-up procedure, making the method more feasible, economical, and environmentally sustainable for the synthesis of thiazoles.

Introduction

The substituted 2-amino thiazole is an important structural moiety in pharmaceutically important precursors [1,2] and can be well recognized as a privileged structure in medicinal chemistry [3]. Cyclization of thiourea with substituted phenacyl bromide is the straightforward and most widely employed approach for synthesizing thiazoles. Recently, the use of heterogeneous catalysts has attracted significant attention in synthetic organic chemistry due to several advantages, including simplicity in operational procedure, non-toxicity, reusability, prolonged life, low cost, and ease of separation upon completion of the reaction. Amberlyst R 21 solid base resin has emerged as an effective heterogeneous catalyst for various organic transformations [4,5]. In this regard, several catalysts have been reported, which include the use of microwave irradiation [6], 2-hydroxypropyl-β-cyclodextrin [7], diammonium hydrogen phosphate [8], copper acetate [9], PEG-sulfonyl chloride resin [10], nanocatalysts [11], ionic liquids [12], MNPs@

*Corresponding Author: S.U. Tekale; E-mail: tekale.sunil@rediffmail.com

SiO_2-Pr-AP [13], Cu-NP/C [14,15], etc. In the present work, an efficient method has been developed for the simple synthesis of thiazoles from variously substituted phenacyl bromides and thiourea using Amberlyst solid base resin as a reusable heterogeneous catalyst (Scheme 1).



Scheme 1: Environmentally benign synthesis of 2-amino-4-substituted thiazole derivatives

The reaction proceeded smoothly, and the products were obtained in high yields using Amberlyst resin in ethanol. To explore the scope of the method, different substituted phenacyl bromides were reacted with thiourea in the presence of 10 wt% Amberlyst resin as the catalyst in refluxing ethanol. After the completion of the reaction, the products were isolated from the reaction mixture, dried, and finally crystallized with hot ethanol. The products were purified by crystallization in ethanol and were characterized by analysis of ^{1}H-NMR, FTIR, and mass spectra.

Materials and Methods

All the chemicals and reagents were from Sigma or SD Fine, which were pure enough that they did not require any further purification before use. The progress of the reaction was observed by Thin Layer Chromatography (TLC) in 30% ethyl acetate: n-hexane on silica gel precoated on aluminum foil. Melting points of the products were recorded in capillaries open at one end on a digital melting point apparatus (Optics Technology) and were uncorrected. The products were purified by crystallization in ethanol, dried, and characterized by analysis of ^{1}H-NMR, FTIR, and mass spectra. ^{1}H-NMR spectra were scanned on a Bruker Avance 500 MHz spectrometer in DMSO-d_6 as the solvent. Chemical shifts were measured in δ ppm downfield with reference to tetramethylsilane (TMS) as the internal standard. FTIR spectra were scanned on a Bruker FTIR spectrophotometer using KBr pellets.

General Procedure for the Substituted 2-Amino-4-Substituted Thiazole Derivatives

In a round-bottom flask, a mixture of phenacyl bromide (1 mmol), thiourea (1.2 mmol), and Amberlyst resin (10 wt%) as a catalyst was added to 1 mL of ethanol, and the contents were refluxed at 80 °C and the reaction progress was monitored by thin-layer chromatography using ethyl acetate: n-hexane (3:7) as the mobile phase. After completion of the reaction in 1 h, the reaction mixture was filtered to isolate the catalyst and then added onto crushed ice to obtain a solid product. Amberlyst resin remained at the bottom of the reaction mixture and could be easily separated, isolated, and reused upon drying in an electric oven at 120 °C. The separated crude product was purified by recrystallization using hot ethanol. Other derivatives of this series were prepared, and the data are summarized in Table 1.

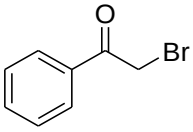
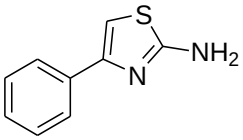
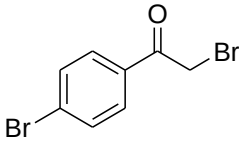
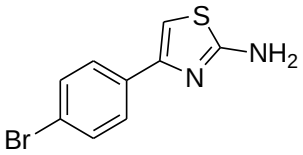
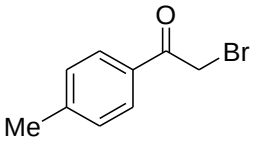
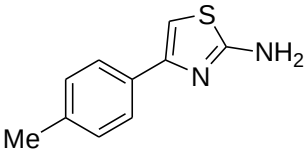
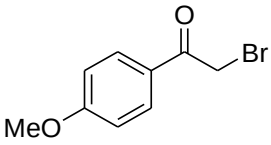
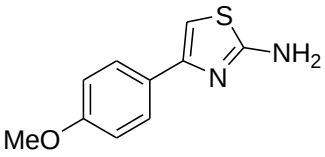
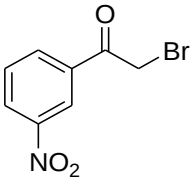
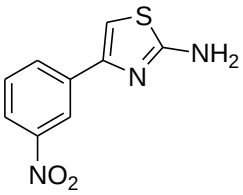
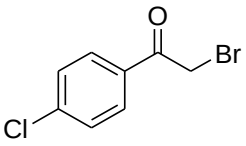
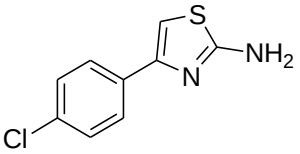
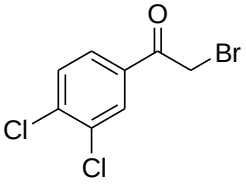
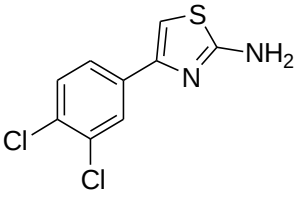
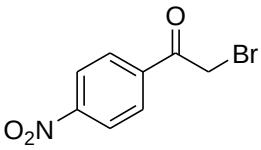
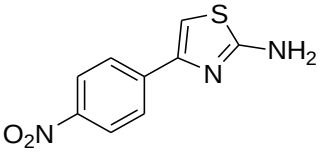
Spectral data of the representative compounds are mentioned below:

4-(4-Bromophenyl)thiazol-2-amine (TU-2): White solid; ^{1}H-NMR (DMSO-d_6): δ ppm 5.82 (s, 2H), 7.33 (d, 1H), 7.71 (d, 3H), 8.74 (s, 1H); IR (KBr) cm^{-1}: 3,142, 2,965, 1,597, 1,070, and 829. Mass: ESIMS =256.9 $(M+1)^+$.

4-p-Tolylthiazol-2-amine (TU-3): White solid; ^{1}H-NMR (DMSO-d_6): δ ppm 2.34 (s, 3H), 7.18 (s, 1H), 7.31 (d, 2H), 7.67 (d, 2H), 9.76 (s, 1H); IR (KBr): cm^{-1} 3,134, 2,837, 1,613, 1,516, 1,321, 1,076, and 821; Mass: ESIMS = 191.0 $(M+1)^+$.

4-(4-Nitrophenyl)thiazol-2-amine (TU-8): Yellow solid, data: ^{1}H-NMR (DMSO-d_6): δ ppm 7.25 (d, 2H), 7.40 (d, 1H), 8.06 (d, 2H), 8.23 (d, 2H); IR (KBr): cm^{-1} 3,107, 2,856, 1,599, 1,340, 1,102, and 849; Mass: ESIMS =222.0 $(M+1)^+$.

Table 1: Synthesis of 2-amino-4-substituted thiazole derivatives using Amberlyst catalyst

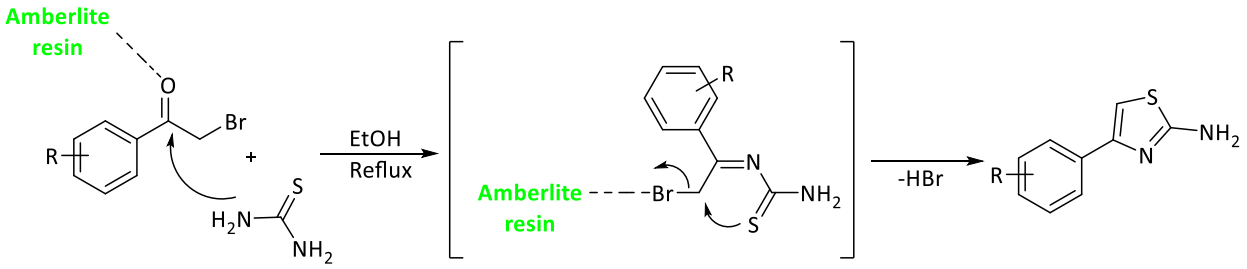
Sr./ No.	Code	R	Product	Yield (%)	M.P. (°C) found	M.P. (°C) reported
1	TU-1			78	147-149	150-151 [13]
2	TU-2			88	222-224	227-230 [13]
3	TU-3			82	136-137	136-138 [14]
4	TU-4			85	204-208	200-202 [14]
5	TU-5			91	279-281	282-283 [13]
6	TU-6			89	203-205	207-208 [13]
7	TU-7			86	190-192	193-194 [15]
8	TU-8			92	280-282	282-284 [14]

Yields isolated in case of reaction of phenacyl bromide (1mmol), thiourea (1.2 mmol), and Amberlyst catalyst (10 wt%).

Results and Discussion

The reaction works well by using Amberlyst resin as a catalyst in ethanol (Table 1). The reaction affords high yields of the corresponding products. The best results were obtained in ethanol as a solvent at 80 °C, making it an environmentally benign and the most suitable solvent. This developed method offers several advantages, including a short reaction time, high yield, operational simplicity, easy workup, and a simple purification procedure.

A possible mechanism for the synthesis of 2-amino-4-substituted thiazole derivatives by the reaction of thiourea and phenacyl bromides catalyzed by the Amberlyst resin as a heterogeneous catalyst is outlined in Scheme 2.



Scheme 2: Mechanism for the synthesis of 2-amino-4-substituted thiazole derivatives

Conclusion

The present method is useful for a rapid and efficient synthesis of 2-amino-4-phenyl-substituted thiazole derivatives in excellent yields. The transformation was found to be successful with different phenacyl bromides possessing electron-donating as well as electron-withdrawing substituents. Considering the pharmaceutical significance of the 2-aminothiazole moiety and growing demand for the development of green and sustainable methodologies, the present approach provides a valuable alternative to existing methods for the synthesis of biologically relevant thiazole derivatives. The present protocol affords several advantages, such as short reaction time, clean reaction profile, easy work-up procedure, and eco-friendly nature, and is a valuable addition to the literature methods for the synthesis of thiazoles.

Acknowledgements

The authors would like to thank the Director, Sophisticated Analytical Instrumentation Facility, Punjab University, Chandigarh, Punjab, and the Director, CSIR-CDRI, Lucknow, for providing spectral analysis. The would also like to express their sincere thanks to the Principal, Deogiri College, Chhatrapati Sambhajinagar, for providing the necessary laboratory facilities.

References

[1] Das, D., Sikdar, P., Bairagi, M. Recent developments of 2-aminothiazoles in medicinal chemistry. *European Journal of Medicinal Chemistry*, **2016**, 109, 89-98.

[2] Alizadeh, S.R., Hashemi, S.M. Development and therapeutic potential of 2-aminothiazole derivatives in anticancer drug discovery. *Medicinal Chemistry Research*, **2021**, 30(4), 771-806.

[3] Chhabria, M.T., Patel, S.V., Modi, P., Brahmkshatriya, P.S. Thiazole: A review on chemistry, synthesis and therapeutic importance of its derivatives. *Current Topics in Medicinal Chemistry*, **2016**, 16, 2841-2862.

[4] Pal, R., Sarkar, T., Khasnobis, S. Amberlyst-15 in organic synthesis. *ARKIVOC: Online Journal of Organic Chemistry*, **2012**.

[5] Tekale, S.U., Kauthale, S.S., Pawar, R.P. Solvent free n-boc protection of amines using amberlyst ra 21 solid base resin as a reusable heterogeneous catalyst. *Journal of the Chilean Chemical Society*, **2013**, 58(1), 1619-1623.

[6] Mamidala, S., Mudigunda, V.S., Peddi, S.R., Bokara, K.K., Manga, V., Vedula, R.R. Design and synthesis of new thiazoles by microwave-assisted method: evaluation as an anti-breast cancer agents and molecular docking studies. *Synthetic Communications*, **2020**, 50(16), 2488-2501.

[7] Zhang, J., Li, P., Zeng, H., Huang, Y., Hong, W. Highly efficient and green synthesis of 2, 4-diphenyl

substituted thiazoles. *Synthetic Communications*, **2020**, 50(5), 735-741.
[8] Balalaie, S., Nikoo, S., Haddadi, S. Aqueous-phase synthesis of 2-aminothiazole and 2-iminothiazolidine derivatives catalyzed by diammonium hydrogen phosphate and DABCO. *Synthetic Communications*, **2008**, 38(15), 2521-2528.
[9] Meshram, H., Kumar, D.A., Vara Prasad, B.R. Copper acetate–catalyzed, mild, highly efficient, and practical synthesis of thiazoles and aminothiazoles. *Synthetic Communications*, **2009**, 39(13), 2317-2320.
[10] Liu, X.L., Wang, Q.Y., Sheng, S.R., Xu, C., Cai, M.Z. Liquid-Phase Organic Synthesis of Thiazoles Using Poly (ethylene glycol)–Bound Sulfonyl Chloride Resin. *Synthetic Communications*, **2008**, 38(19), 3338-3345.
[11] Ji, T., Ye, J., Tang, L., Mandal, T. Recent reports on nanocatalysts in the synthesis of biological thiazoles. *Synthetic Communications*, **2021**, 51(11), 1636-1655.
[12] Potewar, T.M., Ingale, S.A., Srinivasan, K.V. Efficient synthesis of 2, 4-disubstituted thiazoles using ionic liquid under ambient conditions: a practical approach towards the synthesis of Fanetizole. *Tetrahedron*, **2007**, 63(45), 11066-11069.
[13] Ghorbani-Vaghei, R., Alavinia, S., Merati, Z., Izadkhah, V. MNPs@ SiO_2-Pr-AP: A new catalyst for the synthesis of 2-amino-4-aryl thiazole derivatives. *Applied Organometallic Chemistry*, **2018**, 32(3), e4127.
[14] Dhirbassi, A.V., Kauthale, S.S., Tangade, A.D., Kótai, L., Pawar, R.P., Tekale, S.U. Synthesis of 2-amino-4-aryl thiazoles using cu-np/c heterogeneous catalyst. *Organic Preparations and Procedures International*, **2025**, 57(6), 542-548.
[15] Gaikwad, S.A., Patil, A.A., Deshmukh, M.B. An efficient, uncatalyzed, and rapid synthesis of thiazoles and aminothiazoles under microwave irradiation and investigation of their biological activity. *Phosphorus, Sulfur, and Silicon*, **2009**, 185(1), 103-109.

Copyright © 2026 by SPC (Sami Publishing Company) + is an open access article distributed under the Creative Commons Attribution License (CC BY), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.